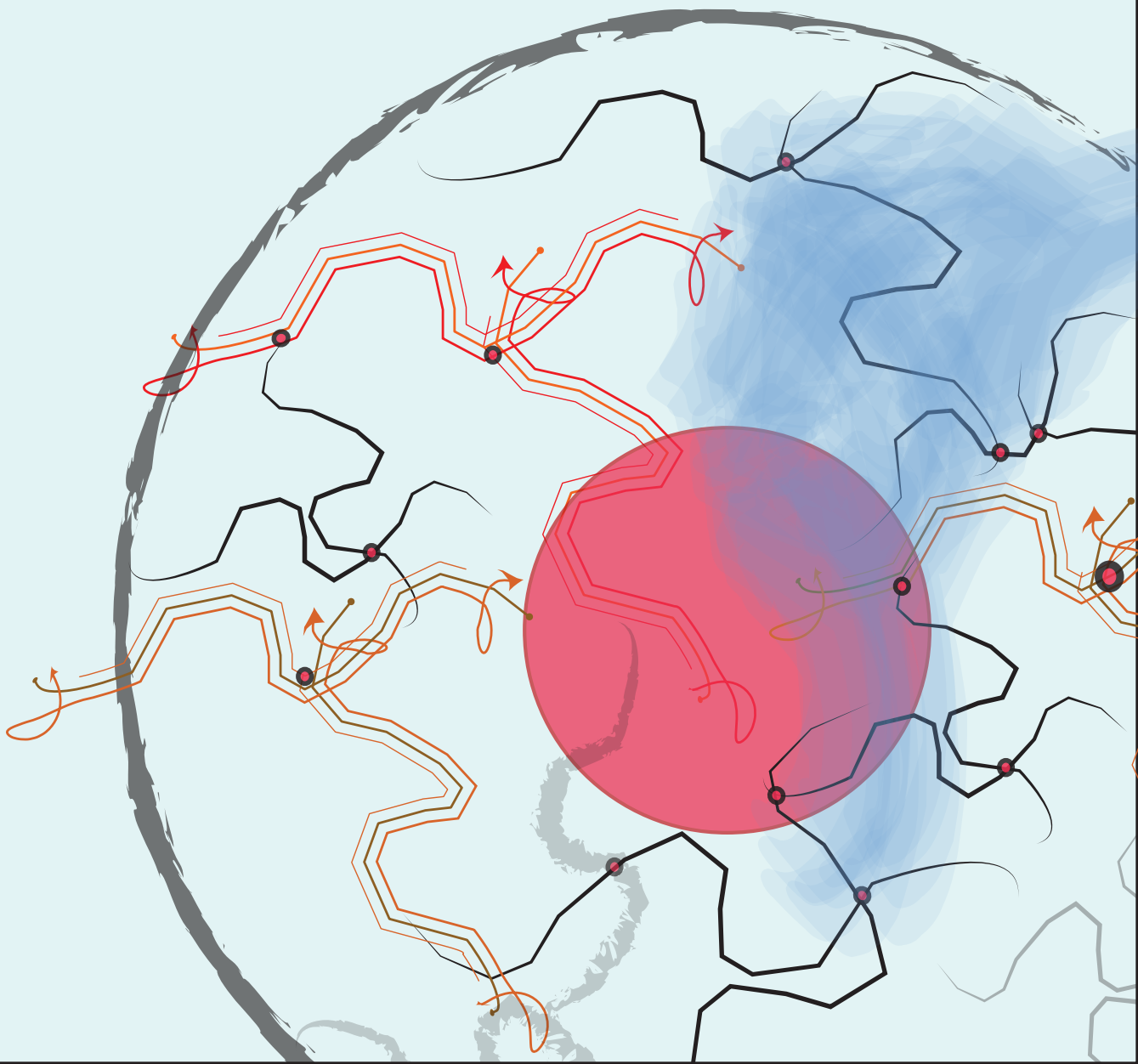
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The role of design in shaping sustainable futures

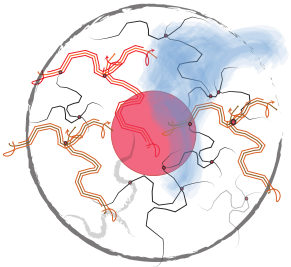
The 14th Design Thinking Research Symposium



DTRS 14

The role of design in shaping sustainable futures

The 14th Design Thinking Research Symposium



Cover image:

An illustration of how a complex design problem can relate to and be interdependent with other things and systems, with intersecting paths and nodes, by Ulrika Florin [1].

[1] Florin, U. (2022) *On design dialogues: Their roots, features, and usage* p 127. in Different Perspectives in Design Thinking, Eriksson, Y. (Ed.).

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Redefining household activities: a metaphorical approach to energy for shaping sustainable futures

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ABSTRACT

People's increased demand to use more land, water, and energy resources with the ways of using them threaten energy resources and increase climate change, but the developments that aim to solve energy and climate crises have so far been technology oriented. There is need to shift away from tech-focused future visions and include more human aspects into planned developments. Therefore, this study explored the ways of enabling people to think outside of the dominant future narratives through two main activities an interactive conference poster and a co-creative workshop. Both activities focused on people's future imaginaries and understanding its interplay with the energy-related household activities, appliances, and the need of sustainable future. The findings show that the use of metaphors can stimulate alternative future imaginaries and enable people to perceive energy-related household activities differently, hence change their consideration while performing such activities. Thus, creating the bridges between humans, their imaginaries on future's household environment and their tools that are used to perform everyday household energy-related activities can help to mitigate energy-related crises and attain sustainable future objective rather than implementing new technological developments to solve energy-related issues.

Keywords: Energy demand, speculative design, metaphors, future imaginaries, co-creation

Introduction

Background

This paper investigates the interplay between residential energy consumption, human aspects, alternative futures, and technological development by exploring how the process of designing household spaces can be a leading tool towards sustainable future. Considering the need for global energy efficiency (Sorrell, 2015), this research endeavor embraces a methodological approach that incorporates speculative design (Dunne and Raby, 2013) within the context of smart and energy-efficient homes. It reflects on how imaginative futures and sustainability principles can inform and transform the process of designing homes, and what impact this might have in relation to reduced environmental impact and enhanced quality of life.

Thus, it starts with delineating current state of energy demand and how it relates to human activities within residential environments which are increasingly surrounded by

technology. Then, it continues to frame how people predominantly deal with the future, how they envision and describe it. After providing a background information about related concept, this paper explores how individuals conceptualize the concept of future's household spaces. The study unfolds through two principal endeavors: the exhibition of an interactive conference poster which designed to engage participants in discussion about future's household appliances, and a co-creative workshop that empowers participants to collectively envision and design the future's living spaces by focusing particularly on the energy challenges while maintaining human comfort. In the light of these two endeavors, the paper aims to contribute to a broader discourse on the necessity of integrating sustainable principles into the daily household activities through highlighting the role of speculative design as a catalyst for change towards sustainable future.

Energy demand and smart solutions

People's social practices, routinized type of behaviors as everyday doings, are interwoven with energy consumption, so every country demands a certain amount of energy to guarantee that people get what they need in their daily lives (Shove and Walker, 2014). In 2018, the global energy consumption increased almost double the average rate of growth since 2010 and continued this upward trend in 2021 (IEA, 2019, 2021). However, increasing global energy demand causes energy scarcities and heavy environmental impact such as global warming (Pérez-Lombard, Ortiz and Pout, 2008). Even from the 1970s, the need of sustainable energy transition and the ways of attaining large-scale societal transformation has been discussed (Loorbach, Frantzeskaki and Avelino, 2017), but energy and climate crisis are more pressing today than ever before. Nowadays many countries consider reducing energy demand, lowering carbon emissions and hence being sustainable as the most urgent and major targets (United Nations, 2019, 2021). Their common goal is mitigating climate change as fast, cheap, and safe as possible through promising ways such as improving energy efficiency and reducing global energy demand (Sorrell, 2015). On the other hand, people spend much more time indoors compared to previous years, and their behaviors and comfort preferences increase energy consumption in households (Klein *et al.*, 2012; Ortiz, Kurvers and Bluyssen, 2017; Bavaresco *et al.*, 2021). In addition, household energy consumption is responsible for 30-38 % of the total energy consumption in different European and American countries (Zhou and Yang, 2016). When people shower, lighten their buildings, cook, or watch TV, they consume energy, but people are not always aware of how much energy is used for different things. Since those activities are part of people's daily behaviors and habits, energy consumption is considered as a by-product of those behaviors and becomes covert (Schwartz *et al.*, 2015). There are some solutions to deal with increasing energy consumption such as informative campaigns to raise awareness or economic incentives or tech-based solutions.

At this point, smart energy systems and smart home technologies have emerged to cease household energy consumption, and recent years have seen many smart technological

innovations (Marikyan, Papagiannidis and Alamanos, 2019; Sovacool and Furszyfer Del Rio, 2020). Smart home technologies (SHTs) refer to digitally enhanced and interconnected systems that observe residents' home behaviors to respond to their comfort and security needs and adjust these systems' responses by forecasting (Aldrich, 2003; Harper, 2011). In addition, SHTs are considered as a core element to discuss energy efficiency and support Europe's strategic net-zero energy technologies (Sovacool and Furszyfer Del Rio, 2020; United Nations, 2021). However, 'smarter' home environments do not always mean more sustainable or better ways of living (Hilty, 2013). Apart from its so-called smartness, energy efficient SHTs bring some risks and barriers that affect people's energy-related behaviors. For example, Sovacool and Furszyfer Del Rio (2020) highlight people's hesitations and concerns about SHTs such as data privacy, technical reliability and precision, usability and learning, energy rebounds and wasteful consumption, as well as loss of personal control and home ownership. Moreover, people seem to prioritize their comfort levels rather than energy efficient considerations since their comfort zones are part of their daily routines in their homes (Kapedani *et al.*, 2022). Thus, technical approaches and high-tech solutions alone will not be sufficient to attain either smart home environments or sustainable behaviors (Rodgers, Bartram and Woodbury, 2011; Cash, Hartlev and Durazo, 2017).

The issue of tech-oriented solutions

As Loorbach et al. (2017) highlight, energy transition does not only refer to a technological change or improvement but also refers to a socio-cultural change that comprises routines and beliefs. However, the scholars and industry solutions have primarily been focused on technical aspects of SHTs rather than the human aspects (Marikyan, Papagiannidis and Alamanos, 2019; Sovacool and Furszyfer Del Rio, 2020). In addition, the developments that aim to solve energy and climate crises have so far been technology- and economy oriented (Loorbach, Frantzeskaki and Avelino, 2017; Marikyan, Papagiannidis and Alamanos, 2019). Hence, the role of humans is mainly neglected or their interactions with developed technologies are not considered thoroughly, and as a consequence developed technologies usually fail or become insufficient (Rodgers, Bartram and Woodbury, 2011; Sovacool and Furszyfer Del Rio, 2020; Bremer *et al.*, 2023). Based on these different theories about tech-oriented developments, this paper draws attention to lack of human aspects in the transition to sustainable energy and the future. Therefore, if humans are considered as subjects, then technological developments can be the tool that humans utilize in everyday life to achieve their sustainable future objectives, see Fig. 1. People's future can be said to be trapped in a vicious circle where technological developments for a sustainable future are mismatched with people's future aspirations. The study presented in this paper explores if these mismatches arise from a lack of understanding about what people envision for their future's life and how they describe what they imagine. Therefore, this study focuses on what people imagine about future's household appliances and energy-related activities by understanding how design can help them to describe their future

imaginaries. Before sharing more details about the study, there is need for clarification how 'the future' is handled by humans in the following section.

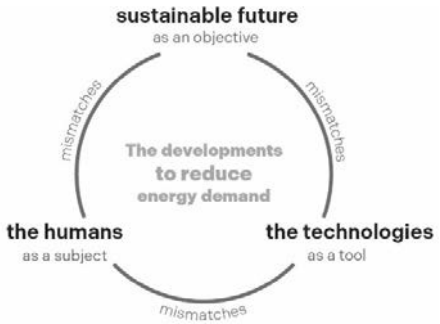


Figure 1: The mismatches between the humans, their sustainable future goals, and the technological development that prevent reducing people's energy demand. Adapted from Selvefors' (2017) activity-system model.

Future Studies and Design

People are generally curious about what the future will bring (and how it could be known) and eager to adjust their future paths in accordance with their plans (how it could be changed) (Höjer and Mattsson, 2000). However, the main problem with the future is that it does not exist yet (Hancock and Bezold, 1994), and any kind of visualizations or descriptions of the future could be considered incomplete or ill-defined (wicked) (Lockton and Candy, 2018) since no one can say if they are true or not. Many organizations and scholars have utilized future studies by generating alternative future scenarios and disrupting dominant way of thinking and making sense of the world to address the ambiguity of the future, anticipate the possibilities, and attain the desired alternative future (Inayatullah, 2004, 2008). Although future studies can support people in creating alternative future scenarios, those scenarios can be found as "flatland of futures" due to its insufficient exploration (Schultz, Crews and Lum, 2012, pp. 129).

On the other hand, the 'wicked' nature of the future aligns with the field of design, as many design situations are also characterized as wicked problems due to their unclear definitions, conditions, and goals, and with their many probable solutions (Rittel and Webber, 1973). In addition, the design field and its inherently future-oriented feature guide individuals and organizations to explore, analyze, and form the future through exploring 'what if'-questions (Binder, Brandt and Gregory, 2008). Speculative design is said to bring more diverse alternative scenarios, through the practice of imagining, asking what-if questions, and generating alternative scenarios for both the current and the future (Auger, 2013). Whereas Wangel, Broms and Runberger (2021) frame speculative design as an umbrella term for all design approaches that explore alternative past, presents and futures, Dunne and Raby (2013 p. 189) describe speculative design as "/…/ a catalyst for social dreaming". This collective mindset indicates the participatory and co-creative approach that can enrich what-if scenarios (Farias, Bendor and Van Eekelen, 2022). Thus, the interaction between design and future studies can bring

richness and multidimensions to the flatland of futures (Angheloiu, Chaudhuri and Sheldrick, 2017), and provide new insights to both fields. Fry (2015, p. 8), for example, states that the importance of design for the future is that "the future with a future for 'us' can only be reached by design". In the same way, the United Nation's Bruntland Commission frames sustainability as a future for people, for 'us', where both the needs of the present and the future are considered without compromising one another (United Nations, 1987). As sustainability is also a future-oriented field like design, the relation between design and the future indicates that design can play a key role for sustainability.

In addition, Pohl and Hirsch Hadorn (2007) describe how to achieve sustainable energy transition in three steps depending on gaining system, target, and transformation knowledge. Whereas system knowledge is about understanding current systems' issues which need to be changed, target knowledge focus on desirable and plausible alternative futures (Pohl and Hirsch Hadorn, 2007; Gaziulusoy and Ryan, 2017). In transformation knowledge, there is need for potential paths that become an exploratory tool to understand how to transform current, undesirable, and unsustainable systems to more desirable and sustainable future systems (Pohl and Hirsch Hadorn, 2007; Gaziulusoy and Ryan, 2017). As emphasized within these steps, people's current issues - and especially their desired future imaginaries and how to make them real - are by these authors seen as crucial for attaining sustainable societal and energy transformation. Lockton and Candy (2018) describe the role of future imaginaries as helping to sensitize people to how they imagine their living system – with its own dynamics – and supporting people to reveal their expectations and aspirations. In the same way, Pohl and Hirsch Hadorn (2007) define the role of future imaginaries as a tool to gain target knowledge and a necessary step to achieve sustainable societal transformation. The crucial role of people's future imaginaries leads this study to focus more on system and target knowledge for exploring how design can help people to envision and describe their alternative and desired futures.

Metaphors as a tool for envisioning

The stated crucial role of people's future imaginaries raises the question of how people can describe their imaginaries around future's smart homes. The use of metaphors can, for this reason, be helpful, as when people reflect on the future especially in relation to complex and unknown or non-existent issues such as sustainable societal transformations. Metaphors is a design method that refers to a way of thinking and talking about something in terms of something else (Lakoff and Johnson, 1980), with the idea of this promoting designers to be thinking outside of the famous box concept with 'the box' being a metaphor in itself. People often use metaphors in their everyday life (Lakoff and Johnson, 1980), even though they might not realize it, and thus it can be seen as an unconscious habit. As Lakoff and Johnson, (1980) highlight, people are surrounded by abstract concepts, such as time, power, and space, which they tend to treat metaphorically. Metaphors are not only about the use of one thing for describing another (Lakoff and Johnson, 1980), but also about making sense of things in a fictional way. The

use of metaphor can enable people to describe their 'wicked' future imaginaries and then transmit their conceptions to someone else. Hence, the metaphorical prospects of design can become a tool for people to express their future conceptions since metaphors are said to support a reframing of the problem and 'thinking outside the box'. Metaphors are said to, for example, bring new perspectives and solutions to the problem (Hey and Agogino, 2008). As design problems are usually described as 'wicked' - and designers are said to approach them metaphorically – this design method has been and can be used for generating and introducing novel ideas during the design process (Alves-Oliveira *et al.*, 2021). People can envision and understand the abstract everyday concepts by using concrete objects. For example, the use of words such as 'desktop' and 'windows' that has helped people transcend from familiar analogue and material items to digital working platforms (Alves-Oliveira *et al.*, 2021).

In addition, when people talk about the future, they are also describing their own or their socially shared future imaginaries (Dunne and Raby, 2013). As their future imaginaries are generally abstract, they can use metaphors to re-brand one concept they are using now to define another thing in the future. One explanation to this is that people's ideas of the future are said to be grounded on their past and present mindsets (Sanders, 2001), they simply use their present vocabulary and image inventory while speculating about the future. When people use metaphors to think and talk about a non-existent future, metaphors can affect their ways of thinking and approaching the future (Alves-Oliveira *et al.*, 2021). However, if people use conventional (Inayatullah, 2004) and fossilized metaphors (Inayatullah *et al.*, 2016), their future imaginaries will not end up any different from their present. Daily metaphors should, for this reason, be thoroughly analyzed to deepen the understanding of both the present needs and the future aspirations. In addition, rather than re-employing current conventional metaphors, disrupting existing ones and generating new, alternative metaphors can enable creating alternative futures (Inayatullah, 2004).

Methodology

The Study

This study presented in this paper sets out with the intention to explore the interplay between human needs (as subject) while performing energy-related household activities through household technological appliances (as tools) and their future imaginaries on energy-efficient living environment (as an objective). This exploration has been done through the lens of sustainability, future imaginaries, and speculative design. The main approach was encouraging the participants to co-explore and co-create, so this collaborative process enabled participants and me, the researcher, to speculate about the future. The study described here consisted of two main activities: an interactive poster and a co-creative workshop. The interactive poster aimed to gather people's current mindsets (subject) that shape their imaginaries and dominant narratives on the future's household appliances alternatives (tools). The co-creative workshop aimed to provide

understanding current and alternative narratives on energy-related household activities (tools), what people (subject) desire and anticipate from alternative scenarios and how these are related with sustainable future (objective), further described in upcoming sections. Therefore, these two main activities of the study provide an exploration of mismatches between people, technologies that are used for household activities and the need of sustainable future.

The Interactive poster

Firstly, I got the invitation to design a poster for the doctoral consortium at the NORDES (Nordic Design Research) conference, held 11-23 June 2023 in Norrköping, Sweden. A poster can be seen as a relative static object; however, the intention was to use this opportunity to gather material for the study. I therefore designed an interactive poster with the aim of gaining a first understanding of what kind of future imaginaries the conference participants (subject) had about future's household appliances (tools). The intention of this poster was to enable people to interact with it and share their thoughts on post-it notes and to the researcher himself. The poster design was for this reason influenced by a past-moment-future timeline (Sanders, 2001) and future cones (Hancock and Bezold, 1994). The poster has a timeline that starts with past, through present, to future, see Fig. 2.



Figure 2: The image shows the interactive poster that was presented in NORDES 2023 (Photos was taken by author.)

Between the past and present sections, there is a series of 'provocative' images on the evolution of domestic floor cleaning products together with two 'provocative' questions and descriptive texts. The background assumption with the provocative aspects was to trigger/nudge participants into sharing their thoughts and opinions. In addition, the timeline divides the poster into hopes (preferable) section on the top, and fears (unpreferable) section on the bottom. In addition, a future section line is multiplied to three extra lines that represent impossible, plausible, and probable futures both on top and bottom parts. Furthermore, two future imaginaries were pre-attached on the poster as examples and triggers for people to interact with the poster. The intention with the

poster design was, hence, that participants should be able to react directly on the poster and to place their post-it notes with imaginaries on their future alternatives.

In this study, the intention was not only focusing on people's desired and preferable futures (a utopia) but also get a better understanding of their fears or nightmares about the future. As Sanders, (2001) emphasizes, people's memories and experiences form their present mindset and future imaginaries which can consist of their hopes and fears. In total, about 40 people interacted with the poster and the researcher during this activity.

The co-creative workshop

The other activity in this study consists of a co-creation workshop in which participants replaced energy-related household activities with metaphors and speculated on how the dynamics of those activities might change in the future. The aim was to explore current dominant and alternative narratives around energy-related household activities, then how the relation between people (subject) and energy-related household activities (tools) can change through using metaphors to reveal their desires and anticipations by creating alternative scenarios. With this goal, master's students in industrial design engineering were invited to participate and 19 students took part. The set up consisted of them being divided into four smaller groups. As Sanders (2000, p.3) indicates "tools" are needed for participants to "express themselves visually and verbally". In the workshop the participants were provided with tools previously developed by Lockton, Singh, et al. (2019). These included a worksheet for each group and a deck of image cards and concept cards were used. In addition, I created a deck of cards that contained energy-related household activities, see Fig 3. Finally, the participants had access to a generative toolkit including two- and three-dimensional components like colorful papers, wires, Legos, and dolls.

Each group selected an energy-related household activity card that they thought is a current issue and still will be an issue in the future. Then each group's participants filled in the first three steps of the worksheets based on their chosen energy-related activity. Next step was generating new metaphors for describing selected activities, so they picked cards from the image- and concept decks of cards. After following next three steps of the material canvas, they started to think about what kind of alternative scenarios there can be if they use new metaphors for selected energy-related activities. For example, whereas the group who picked charging as an issue came up with 'sealing the leakages' as a new metaphor, another group used 'de-characterization' as a new metaphor rather than saying doing laundry. The rest of the groups generated 'mindful workout sessions' and 'dance of flavors' as new metaphors for cleaning house and cooking, respectively. After assigning new metaphors for the energy-related activities, each group co-created a scene that showed either an environment, appliances, or systems based on the new metaphors created through the generative toolkit. For example, the group who used 'de-characterization' imagined an alternative everyday life

where people can de-characterize their clothes and themselves (perhaps) through a new device which is available for public use and for individual use at home.



Figure 3: The image shows one of the participants groups' worksheet and their model regarding their new metaphor. (Photos was taken by author.)

Findings

The main method for analyzing the activities was thematic analyses, since it is an approach for probing patterns within data to determine, organize and construe them (Braun and Clarke, 2006).

The interactive poster

In the poster design activity, the materials consisting of the participants' outspoken thoughts and the post-it notes were assembled based on the relations to each other. Then, these were summarized into the three overall themes of 'non-human', 'human-centric' and 'doubtful future'. The participants' future visions advocates for more-tech and autonomy were categorized under the 'non-human' theme, since they illustrate a future devoid of human attributes. Since different themes indicate humane aspects like care, fear, and hesitancy, those expressions were gathered under the 'human-centric' theme. The last theme was 'doubtful future' because each of the participants' expressions shows some kind of uncertainty or even almost impossibility. Although the most of future imaginaries on household appliances were about high-tech, another strong and inevitable imaginary is the desires and hopes of human interaction with the household appliances and in the household activities. First indication is that simplicity will be a valued feature in the future. For example, one of the participants' expressed desires which point at the main function of a kettle by saying, "smart but it is just a kettle" (Participant A, June 2023). Also, another participant underpinned the simplicity theme by sharing their vision on "having a household which is still comfy and not too tech" (Participant B, June 2023). The findings indicate that uncontrollable and unprojected situations can cause crisis of routines. For instance, one participant mentioned how covid has changed people's housing. The mainstream can be imagining a

utopia that people will be surrounded by assistive technologies. However, considering recent Ukraine-Russia war, many people suddenly faced various problems and regarding energy aspect they lived with power-outages or cuts, hence limited technology (European Union, 2023). Thus, increased climate crises and resource scarcity seems to be nudging people into also thinking of future scenarios that have limited and unsustainable technologies. From the results of this poster design activity, a finding is that technology integration in people's daily lives is inevitable since people desire more. Finally, not only people's utopias but also dystopias should be considered in future scenarios to explore different perspectives and possibilities.

The co-creative workshop

The co-creation workshop was analyzed through the collected materials (worksheets and new metaphor cards), observations, and the author's notes during the workshop. Firstly, the use of new metaphors enables people to envision alternative ways of perceptions and interactions for energy-related household activities. For example, the use of the metaphor 'sealing the leakages' for the charging activity rearranged the approach to wasting electricity. For this reason, participants thought that people need to turn off their tablets when they are not used for sealing the electricity leakages. This understanding is an example of habit substitution by destabilizing the context (Gardner *et al.*, 2021). Secondly, the use of the metaphor of a 'mindful workout session' for cleaning the house unleashed what could be seen as the problematic interaction between human and cleaning houses, such as tech dependent cleaning, more complicated, more expensive, and new ways of cleaning through new appliances rather than a simple dustcloth and a floor mop. Participants reflected that the perception of the mainstream way of cleaning the house on social media, such as steam cleaner and robot vacuum cleaner, create a social pressure. Instead, they envisioned a 'mindful workout session' that would enable them to take care of both their homes and themselves. This result also refers to the human-centric theme of the interactive poster by emphasizing human effort with an act of care and art. Thus, one finding is that the activity of defining new metaphors seemed to enable people to understand the current issues which need to be changed (system knowledge) and then focus on desirable and plausible alternative futures (target knowledge). Another perspective is that new metaphors bring a re/valuing of the energy-related household activities. With the use of the 'de-characterization' metaphor, participants added new values on doing laundry activities, hence, they reconsidered the meaning of hygiene and its norms. Adding new values enabled them to consider reducing their laundry frequency with the desire of keeping the same characteristics of their clothes, rather than washing (de-characterizing) them.

Discussion

Shove and Walker (2014) highlight that people's daily practices demand more energy. Although humans cause energy crises by demanding more energy, they are also affected from energy-related issues. Thus, this dilemma places humans at the core of the energy

and climate crises. Since the value of doing and being part of something is in human nature, every development that just focuses on technology-driven solutions will miss the human needs and desires for the future and either fail or be insufficient. As some of the participants expressed that high-tech household appliances and solutions can assist people but cannot take the lead for a long term. Thus, some activities need human interactions due to their values, like "the art behind coffee making" as one of the attendants mentioned. Technology integration will not minimize human interaction because human cannot be in an ineffective role. The study presented in this paper shows that a change on how people call and define an energy-related activity like doing laundry or cleaning the house have the possibility to change their perceptions with such energy-related activities. By just calling the activity 'de-characterization' or 'mindful workout sessions' and without adding another tech-development, the participants reconsidered the dominant norms around hygiene and the way of washing their clothes or re-value the effort that is performed while cleaning the house. In addition, the use of 'sealing the leakages' metaphor redefines the activity of charging a device as a waste of resources rather than considering it as refilling the power of a device. This new way of perceiving the activity can change people's attitudes toward more energy-efficient everyday life. An important finding from this study is that there is a mismatch between technological developments, humans' behaviors, and desires, and their sustainable future goals which should be further addressed. The mismatch can be just a misaligned point of view of an individual to the activity itself, i.e., rather than adding another technical solution, the importance is to be understanding how people perceive the present and what they envision for the future. Furthermore, humans are not passive actors who just engage with some technological tools: they are actors who live in the present and dream of a future that they want. As Selvefors (2017) highlight, a subject (a human) utilizes a tool to attain her/his objective. In this case, if humanity aims for a sustainable future and if humanity is surrounded by technology, then the mismatches between those two things needs to be better understood and addressed.

Conclusion

This paper delves into shifting away from technology-oriented future imaginaries through two different ways of encouraging people to envision the future differently. The findings indicate various future alternatives from a scenario where the human features are at the core which values the care and desires doing things by yourself to non-human approach is prioritized where smart appliances dominate chores. In addition, this study contributes to expand the field of design and future research, particularly in the employment of metaphors which seemed to stimulate alternative imaginaries. The use of new metaphors can change how people perceive or value energy-related household activities, so this shift in the thinking about such activities can enable people to both think and act differently. The main importance of this is that the human aspect is crucial to be able to develop an energy-efficient future and the many different ways humans can be included in developing that future together. The current study can be seen as a pre-

study for other activities and studies aimed at better understanding how people think about the future and what methods and means that can be used to make them think about the future to change behaviors in the present. The limitation of the study is that it was a time- and participant limited endeavor, yet the findings indicate some promising areas that needs to be further explored in future studies.

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